Classical

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Single bit Operations

- *An operator which updates the state of* a bit *, depending on the current state.*
- We can apply four different operators to a single bit:
  - Identity: I(0)=0 and I(1)=1
  - Negation: NOT(0)=1 and NOT(1)=0
  - Constant (Zero): ZERO(0)=0 and ZERO(1)=0
  - Constant (One): ONE(0)=1 and ONE(1)=1

$I =$

	$\hookleftarrow$	*initial* states **0**	*initial* states **1**
final states	**0**	1	0
	1	0	1

$NOT =$

	$\hookleftarrow$	*initial* states **0**	*initial* states **1**
final states	**0**	0	1
	1	1	0

$ZERO =$

	$\hookleftarrow$	*initial* states **0**	*initial* states **1**
final states	**0**	1	1
	1	0	0

$ONE =$

	$\hookleftarrow$	*initial* states **0**	*initial* states **1**
final states	**0**	0	0
	1	1	1

$I =$

	$\hookleftarrow$	*initial* **0**	*states* **1**
final	**0**	1	0
states	**1**	0	1

$NOT =$

	$\hookleftarrow$	*initial* **0**	*states* **1**
final	**0**	0	1
states	**1**	1	0

$ONE =$

	$\hookleftarrow$	*initial* **0**	*states* **1**
final	**0**	0	0
states	**1**	1	1

$ZERO =$

	$\hookleftarrow$	*initial* **0**	*states* **1**
final	**0**	1	1
states	**1**	0	0

Reversibility and Irreversibility

- If we can recover the initial value(s) from the final value(s), then the operator is called reversible like Identity and NOT operators.
- If we cannot know the initial value(s) from the final value(s), then the operator is called irreversible like ZERO and ONE operators.
- **This classification is important, as the quantum evolution operators are reversible.**
- The Identity and NOT operators are two fundamental quantum operators.

A Probabilistic State

- Eg : Outcome of Flipping a fair coin

$FairCoin =$

$\hookleftarrow$	**Heads**	**Tails**
Heads	$\frac{1}{2}$	$\frac{1}{2}$
Tails	$\frac{1}{2}$	$\frac{1}{2}$

Biased Coin

- A probabilistic bit is in states 0 and 1 with probabilities p and 1−p, where p is a number between 0 and 1.
- If p=1 or p=0, then the bit becomes deterministic.
- A biased coin is a coin that has been designed or altered in such a way that it does not provide equal chances for each outcome when flipped.
  - You have a coin that lands **heads** with probability 0.3 and **tails** with probability 0.7
  - After a toss the coin isn't *deterministically* in one state—it's in a **probabilistic state**
    - P(state 0)=0.3, P(state 1)=0.7

- If the coin has a bias then our information about the outcome will be as follows:

$$\frac{Pr(Heads)}{Pr(Tails)} = \frac{3}{1}$$

- The outcome will be heads with probability 0.75 and the outcome will be tails with probability 0.25.

- Listing probabilities as a column $\begin{pmatrix} 0.75 \\ 0.25 \end{pmatrix}$.

Exercise : Suppose that Asha flips a coin having the bias $\frac{Pr(Heads)}{Pr(Tails)} = \frac{1}{4}$. Represent your information about the outcome as a column vector.

Vector representation

- Suppose that we have a system with 4 distinguishable states:
  - s1, s2, s3, and s4.
- We expect the system to be in one of them at any moment.
- By using our column representation, we can show each state as a column vector by using the vectors in standard basis of $\mathbb{R}^4$

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \text{ and } e_4 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Vector representation

- This representation helps us to state our information on a system when it is in more than one state with certain probabilities.
- For example, suppose that the system is in states s1, s2, s3, and s4 with probabilities 0.20, 0.25, 0.40, and 0.15, respectively.
- *The total probability must be 1* 0.20+0.25+0.40+0.15=1.0
- Then, we can say that the system is in the following probabilistic state:

$$0.20 \cdot e_1 + 0.25 \cdot e_2 + 0.40 \cdot e_3 + 0.15 \cdot e_4$$

$$= 0.20 \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + 0.25 \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + 0.40 \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + 0.15 \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0.20 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0.25 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0.40 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0.15 \end{pmatrix} = \begin{pmatrix} 0.20 \\ 0.25 \\ 0.40 \\ 0.15 \end{pmatrix},$$

where the summation of entries must be 1.

- A probabilistic state is a linear combination of the vectors in the standard basis.
- Here coefficients (scalars) must satisfy certain properties:
  - Each coefficient is non-negative
  - The summation of coefficients is 1
- Alternatively, we can say that a probabilistic state is a probability distribution over deterministic states.
- We can show all information as a single mathematical object, which is called as a stochastic vector.

- ***Remark that the state of any linear system is a linear combination of the vectors in the basis.***

You have a coin that lands **heads** with probability 0.3 and **tails** with probability 0.7

- Case 1 : Tossing the bias coin once
  - P(Head) = ? , P(Tail) = ?

- Case 2 : Tossing the bias coin twice
  - P(Head) = ? , P(Tail) = ?

Case 3 : Our friend Asha has one euro and one cent.

- Both coins are biased, and the probabilities of getting heads and tails are as follows:
  - one euro: heads with probability 0.6 and tails with probability 0.4.
  - one cent: heads with probability 0.3 and tails with probability 0.7.

What is the joint probability of getting HH ,HT ,TH and TT (First one Euro and second one cent)

Case 4 : Our friend Asha has one euro and one cent.

- Both coins are biased, and the probabilities of getting heads and tails are as follows:
  - One euro: heads with probability 0.6 and tails with probability 0.4.
  - One cent: heads with probability 0.3 and tails with probability 0.7.

What is the transition probability if Asha flips her coins based on the following **protocol**:

- she starts with flipping one euro(fixed)
  - whenever she gets heads, she flips one euro in the next round and,
  - whenever she gets tails, she flips one cent in the next round.

Probabilistic Operators

$$GameCoins = \begin{array}{c|cc} \hookleftarrow & \textbf{Head} & \textbf{Tail} \\ \hline \textbf{Head} & 0.6 & 0.3 \\ \textbf{Tail} & 0.4 & 0.7 \end{array} \equiv \begin{array}{c|cc} \hookleftarrow & \textbf{0} & \textbf{1} \\ \hline \textbf{0} & 0.6 & 0.3 \\ \textbf{1} & 0.4 & 0.7 \end{array}$$

She starts in state 0: $v_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

State 0 represents Head and state 1 represents Tail.

After first coin flip:

$$\left(GameCoins = \begin{array}{c|cc} \hookleftarrow & \textbf{Head} & \textbf{Tail} \\ \hline \textbf{Head} & 0.6 & 0.3 \\ \textbf{Tail} & 0.4 & 0.7 \end{array} \right) \left(CurrentState = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right)$$

Each entry of the new vector is calculated by a row and current state:

$$\begin{pmatrix} 0.6 \cdot 1 \\ + \\ 0.3 \cdot 0 \\ \hline 0.4 \cdot 1 \\ + \\ 0.7 \cdot 0 \end{pmatrix} = \begin{pmatrix} 0.6 + 0 \\ 0.4 + 0 \end{pmatrix} = \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix}.$$

What is the state after second coin flip ?

After second coin flip:

$$\left(GameCoins = \begin{array}{c|cc} \hookleftarrow & \textbf{Head} & \textbf{Tail} \\ \hline \textbf{Head} & 0.6 & 0.3 \\ \textbf{Tail} & 0.4 & 0.7 \end{array} \right) \left(CurrentState = \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix} \right)$$

Each entry of the new vector is calculated by a row and current state:

$$\begin{pmatrix} 0.6 \cdot 0.6 \\ + \\ 0.3 \cdot 0.4 \\ \hline 0.4 \cdot 0.6 \\ + \\ 0.7 \cdot 0.4 \end{pmatrix} = \begin{pmatrix} 0.36 + 0.12 \\ 0.24 + 0.28 \end{pmatrix} = \begin{pmatrix} 0.48 \\ 0.52 \end{pmatrix}.$$

Coin-flipping protocol of Asja is a *probabilistic operator*.

Similarly to any operator, depending on the current state, Asja's coin-flipping protocol determines the next state.

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \xrightarrow{\text{Asja's coin-flipping protocol}} \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix} \xrightarrow{\text{Asja's coin-flipping protocol}} \begin{pmatrix} 0.48 \\ 0.52 \end{pmatrix}.$$

A probabilistic operator evolves the system from a probabilistic state to a probabilistic state.

- A probabilistic operator can be represented as a square table or matrix.
- The entries of a probabilistic operator represent the transition probabilities between the states.
- Therefore, **each entry is non-negative.**
- Each column represents the transition probabilities from a state to all states. Therefore, **the summation of all entries in each column is 1**, i.e., probability 1 is distributed over all states.
- Any matrix satisfying these two properties is called a **stochastic matrix**.
- A probabilistic operator is a stochastic matrix, and vice versa.
- *Remark that the operator of any linear system is represented as a matrix.*

Probabilistic evolution

- A probabilistic state is a stochastic vector, say v.
- A probabilistic operator is a stochastic matrix, say A,
- If probabilistic operator A is applied to probabilistic state v, the new state, say v′, is calculated as

$$v' = A \cdot v.$$

- *The evolution of linear system is represented by matrix-vector multiplication.*

- *Exercise :* The operator $GameCoins = \begin{pmatrix} 0.6 & 0.3 \\ 0.4 & 0.7 \end{pmatrix}$ is applied to the probabilistic state $\begin{pmatrix} 0.1 \\ 0.9 \end{pmatrix}$.

Representation for states 0 and 1

The vector representation of state 0 is $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Similarly, the vector representation of state 1 is $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

We use $\lceil \mathbf{0} \rfloor$ to represent $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\lceil \mathbf{1} \rfloor$ to represent $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Then, the probabilistic state $\begin{pmatrix} 0.2 \\ 0.8 \end{pmatrix}$ is also represented as $\mathbf{0.2}\lceil \mathbf{0} \rfloor + \mathbf{0.8}\lceil \mathbf{1} \rfloor$.

Similarly, the probabilistic state $\begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix}$ is also represented as $\mathbf{0.6}\lceil \mathbf{0} \rfloor + \mathbf{0.4}\lceil \mathbf{1} \rfloor$.

Two Probabilistic Bits

- One single bit as a single system, then what are the states of the system
  - Total 2 states
  - 0 and 1

- If we combine two single bit as a single system, then what is the state of the combined system?
  - In total, we have four different states. We can name them as follows:
    - 00: both bits are in states 0
    - 01: the first bit is in state 0 and the second bit is in state 1
    - 10: the first bit is in state 1 and the second bit is in state 0
    - 11: both bits are in states 1

Suppose that we have two probabilistic bits, and our probabilistic states respectively are $\begin{pmatrix} 0.2 \\ 0.8 \end{pmatrix}$ and $\begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix}$.

Composite systems

When two systems are composed, then their states are tensored to calculate the state of composite system.

The probabilistic state of the first bit is $\begin{pmatrix} 0.2 \\ 0.8 \end{pmatrix} = 0.2\lceil 0\rfloor + 0.8\lceil 1\rfloor$.

The probabilistic state of the second bit is $\begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix} = 0.6\lceil 0\rfloor + 0.4\lceil 1\rfloor$.

Then, the probabilistic state of the composite system is $\left(0.2\lceil 0\rfloor + 0.8\lceil 1\rfloor\right) \otimes \left(0.6\lceil 0\rfloor + 0.4\lceil 1\rfloor\right)$.

Tensor product distributes over addition in the same way as the distribution of multiplication over addition.

$a\lceil 1\rfloor \otimes b\lceil 0\rfloor = (ab)\,(\lceil 1\rfloor \otimes \lceil 0\rfloor) = ab\lceil 10\rfloor$.

The probabilistic state of the composite system is $\left(0.2\lceil 0\rfloor + 0.8\lceil 1\rfloor\right) \otimes \left(0.6\lceil 0\rfloor + 0.4\lceil 1\rfloor\right)$.

$$\left(0.2\lceil 0\rfloor + 0.8\lceil 1\rfloor\right) \otimes \left(0.6\lceil 0\rfloor + 0.4\lceil 1\rfloor\right) =$$

$$0.12\left(\lceil 0\rfloor \otimes \lceil 0\rfloor\right) + 0.08\left(\lceil 0\rfloor \otimes \lceil 1\rfloor\right) + 0.48\left(\lceil 1\rfloor \otimes \lceil 0\rfloor\right) + 0.32\left(\lceil 1\rfloor \otimes \lceil 1\rfloor\right) =$$

$$0.12\lceil 00\rfloor + 0.08\lceil 01\rfloor + 0.48\lceil 10\rfloor + 0.32\lceil 11\rfloor.$$